

MET CS 555 – Foundation of Machine Learning

Module - 2

Lecture - 3: Confidence Interval, Hypotheses Tests, z-test, t-test

Lecture - 4: Scatter Plots, Correlation, Linear Regression

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Statistical Inference

Statistical inference provides methods for drawing conclusions about a population from sample data.

The **goals of statistical** inference are:

- ▷ **Draw conclusion** about a population based on sample data
- ▷ **Provide a statement**, expressed in the language of probability, of how much confidence to be placed in the conclusion

Two of the most common types of **statistical inference** include:

- ▷ **Confidence Intervals** for estimating the value of a population parameter
- ▷ **Tests of Significance** or Hypothesis Tests which assess the evidence for a scientific claim.

Confidence Intervals

Confidence Interval

What is a Confidence Interval?

A **confidence interval** is a range of values that we believe contains the true value of a population parameter (like the average height of all students in a school) with a certain level of confidence (like 95%).

Imagine you're fishing and want to estimate the number of fish in a lake. Instead of counting every fish (which is impossible), you take a sample and use it to estimate the total number. The confidence interval is like saying, "Based on my sample, I believe the total number of fish is between 1,000 and 1,200 with 95% confidence."

Why Do We Use Confidence Intervals?

More Informative than Point Estimates: Instead of just saying "the average height is 170 cm," we provide a range, like "the average height is between 168 cm and 172 cm."

Expresses Uncertainty: It acknowledges that our estimate isn't perfect and shows how much it might vary.

Example - Call Center Calculating Confidence Interval

Suppose a company would like to estimate the average amount of time callers are on hold before they reach a customer service representative.

Executives from the company plan to randomly sample 100 calls and use the sample mean $\bar{x}$, to estimate population mean μ (the average of all callers wait times).

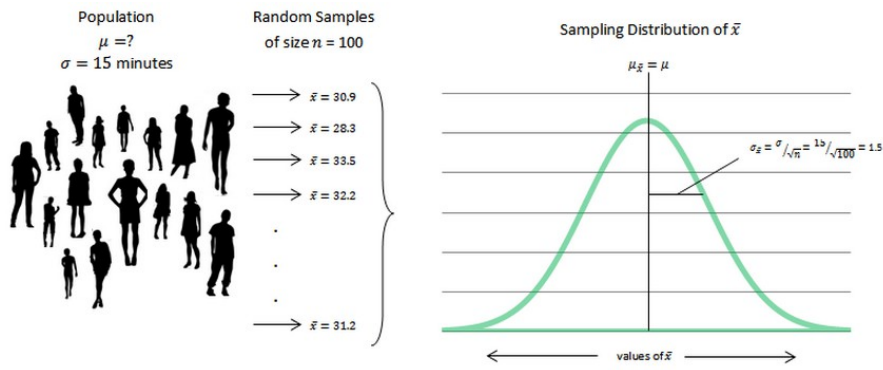
Assume that $\sigma = 15$ minutes and sample mean of the wait times is $\bar{x} = 30.9$ minutes. The parameter of interest is the population mean, μ .

Executives also want to compute a confidence interval to assess the accuracy of the point estimate (the sample mean).

Example calculating confidence interval

Since $n \geq 30$ (here, $n=100$), we know that the sample mean is approximately normally distributed with a mean $\mu_{\bar{x}} = \mu$ and a standard deviation of

$$\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}} = \frac{\sigma}{\sqrt{100}} = 1.5 \text{ minutes.}$$



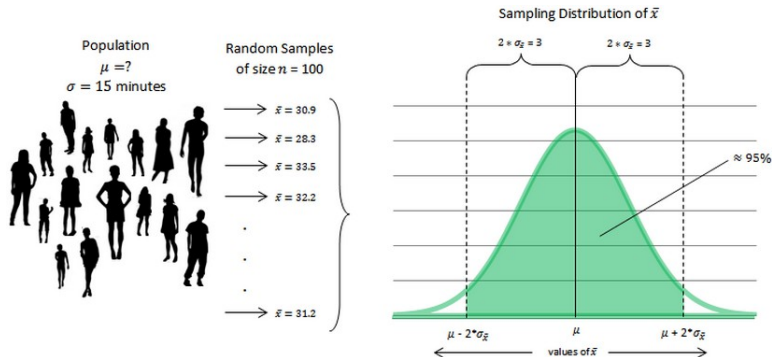
An example calculating confidence interval

In a normal distribution, 95% of the sample means are within 2 SDs of the population mean ($z = 1.96$ corresponds to an area of 95% under the curve). The sample mean, $\bar{x}$, and the population mean, μ , are within three minutes (2 SDs) of each other 95% of the time.

If our estimate of the population mean is between $\bar{x} \pm 3$

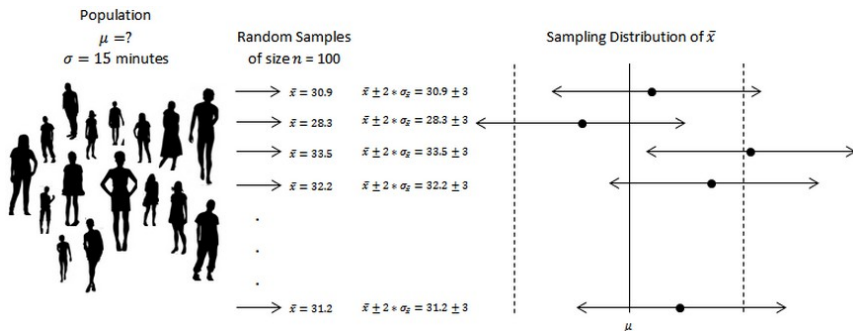
$\bar{x} - 3 = 30.9 - 3 = 27.9$ and

$\bar{x} + 3 = 30.9 + 3 = 33.9$ then we will be right 95% of the time.



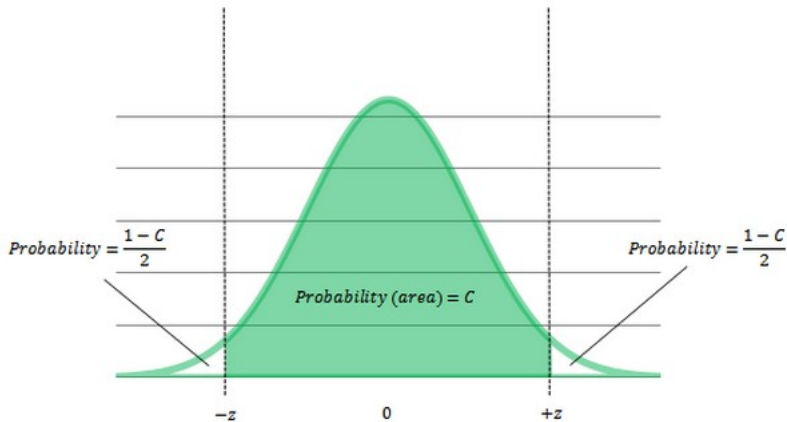
An example calculating confidence interval

- ▶ We know that **95% of the sample means will be within 2 SDs of the population mean.**
- ▶ The sample mean 2 times the SD ($\bar{x} \pm 2 \cdot \sigma_{\bar{x}}$) is called the 95% confidence interval for the population mean.
- ▶ The two-way arrows in the figure represent the confidence intervals from each random sample. The center point is the sample mean and the end of the arrows show the edges of each interval.



Confidence interval

Standard Normal Distribution ($\mu = 0, \sigma = 1$)



Confidence interval

- ▷ Confidence intervals are of the form: **estimate** $\pm$ **margin of error**
- ▷ To calculate a confidence interval with a confidence level of C for the population mean, μ , we use the following formula:

$$\bar{x} \pm z \cdot \frac{\sigma}{\sqrt{n}}$$

- ▷ Generally, 95% confidence intervals are standard. Other common intervals are 90% and 99%.

Confidence Level, C	90%	95%	99%
Critical Value, z	1.645	1.960	2.576

Exercise 1 - Call Center Data Waiting Time

Suppose the executives would like to estimate the average amount of time callers are on hold before they reach a customer service representative by computing a 99% confidence interval. The population standard deviation of wait times, σ , is 15 minutes.

The sample mean of the wait times from the 100 random calls in the sample is 30.9 minutes.

Exercise 1 - Call Center Data Waiting Time

Suppose the executives would like to estimate the average amount of time callers are on hold before they reach a customer service representative by computing a 99% confidence interval. The population standard deviation of wait times, σ , is 15 minutes.

The sample mean of the wait times from the 100 random calls in the sample is 30.9 minutes.

To calculate a confidence interval with a confidence level of C for the population mean, μ , we use the formula: $\bar{x} \pm z \times \frac{\sigma}{\sqrt{n}}$

The 99% confidence interval is calculated as follows:

$$\begin{aligned}\bar{x} \pm z \times \frac{\sigma}{\sqrt{n}} &= \\ 30.9 \pm 2.576 \times \frac{15}{\sqrt{100}} &= \\ 30.9 \pm 2.576 \times 1.5 &= \{27.036, 34.764\}\end{aligned}$$

The Sample size and the **margin of error**

In the case of the confidence interval for the population mean, the margin of error is equal to

$$m = z \times \frac{\sigma}{\sqrt{n}}$$

We like our confidence intervals to be narrow (small in width) with a high amount of confidence. The precision (width) of the confidence interval decreases as:

- ▷ **z gets smaller.** Smaller values of z are associated decreased confidence levels. There is a **trade off between the confidence level and the margin of error.**
- ▷ **σ gets smaller.** When the variability of the individual observations **is reduced**, it becomes easier to estimate the population mean with **higher precision.**
- ▷ **n gets larger.** **Increasing the sample size n reduces the margin of error** for a fixed confidence level C .

The Sample size and the **margin of error**

In the case of the confidence interval for the population mean, the margin of error is equal to

$$m = z \times \frac{\sigma}{\sqrt{n}}$$

To obtain a specific **margin of error**, **m**, for a desired confidence level, **C**, the number of observations needed is:

$$n = \left(\frac{z \times \sigma}{m} \right)^2$$

Exercise 2 - How many callers must they sample?

Suppose the executives would like to estimate the average amount of time callers are on hold before they reach a customer service representative by computing a 95% confidence interval.

As before, the population standard deviation of wait times, σ , is 15 minutes. The executives would like the half width of the confidence interval to be 1.0 minutes (that is, they'd like the margin of error to be 1.0).

How many callers must they sample to get their 95% confidence interval to have the desired width?

Exercise 2 - How many callers must they sample?

How many callers must they sample to get their 95% confidence interval to have the desired width?

To obtain a specific margin of error, m , for a desired confidence level, C , we use the formula to determine the number of observations needed:

$$n = \left(\frac{z \times \sigma}{m} \right)^2 = \left(\frac{1.96 \times 15}{1} \right)^2 = 29.4^2 = 864.36$$

Note: In calculations involving sample size we always **round up (instead of down)** to achieve the requested margin of error. In our example we need to have **samples of 865 callers**

Confidence Level, C	90%	95%	99%
Critical Value, z	1.645	1.960	2.576

Advantages of Confidence Intervals

Provides a Range of Estimates

Offers a plausible range for population parameters instead of a single point estimate.

Example: *Medical Research* – A study reports that a new drug reduces blood pressure by 8-12 mmHg with 95% confidence.

Incorporates Sample Size and Variability

Larger samples yield narrower intervals, indicating more precise estimates.

Example: *Election Polling* – Increasing the sample size from 500 to 1,000 voters narrows the confidence interval from 53%-57% to 54.5%-55.5%.

Facilitates Hypothesis Testing

Helps determine if results are significant and if hypothesized values lie within interval.

Example: *Business Quality Control* – A confidence interval for defect rates (1.2%-2.8%) helps decide if process improvements are needed.

Easy Communication

Simplifies the presentation of statistical findings to non-experts.

Example: *Public Health Reporting* – Reporting average hospital beds as 2.5-3.5 per 1,000 population makes resource planning clear.

Versatile Applications

Broad Use Cases: Applicable to means, proportions, differences between groups, etc.

Example: *Educational Research* – Comparing average test scores between two teaching methods with confidence intervals to assess effectiveness.

Limitations

Misinterpretation Risks

Believing a 95% CI means there's a 95% probability the true parameter lies within the interval.

Fix: Explain that the confidence level refers to the method's reliability over many samples, not the probability for a specific interval.

Dependence on Assumptions

Relies on normality, random sampling, and independence.

Fix: Incorporate steps to verify assumptions before constructing CIs.

Sensitive to Sample Size

Small samples lead to wide intervals; very large samples may produce overly narrow intervals.

Fix: Discuss how sample size affects the margin of error and the usefulness of the interval.

Doesn't Imply Causation

Association vs. Causation: CIs indicate association, not causal relationships.

Fix: Teach the difference between correlation and causation when interpreting CIs.

Common Errors in Confidence Intervals

Using Incorrect Distribution (Z vs. t)

Mistake: Applying Z-scores for small samples with unknown population variance.

Fix: provide guidelines – Use t-distributions when sample sizes are small and population variance is unknown.

Calculation Mistakes

Mistake: Errors in computing standard error or margin of error (e.g., dividing by n instead of $\sqrt{n}$).

Fix: step-by-step practice – Offer detailed procedural exercises and encourage the use of calculators or software for verification.

Confusing CIs with Prediction Intervals

Mistake: Believing CIs predict where individual future observations will lie.

Fix: differentiate concepts – Clearly explain that CIs estimate population parameters, while prediction intervals estimate ranges for individual data points.

Example of Fixing Misinterpretation:

Example: A student interprets a 95% CI of 45%-51% in an election poll as having a 95% chance that Candidate A's support falls within this range for any individual voter.

Correction: Explain that the 95% CI means that if many samples are taken, approximately 95% of the calculated intervals will contain the true population proportion, not that any single interval has a 95% probability of containing the true value.

Hypotheses Tests

Set up the hypotheses

▷ Hypothesis testing is a statistical method used to decide whether there is enough evidence in a sample of data to infer that a certain condition holds true for the entire population.

Why is Hypothesis Testing Important?

- ▷ Decision making: Helps in making informed decisions based on data.
- ▷ Scientific research: Essential for validating theories and experimental results.
- ▷ Business applications: Used in quality control, market research, and more.

Null Hypothesis (H_0)

- ▷ A statement of no effect or no difference. It represents the status quo or a baseline assumption.
- ▷ Serves as the default assumption that there is no relationship between variables.

Alternative Hypothesis (H_a or H_1)

- ▷ A statement that contradicts the null hypothesis. It represents what you aim to support.
- ▷ Suggests that there is an effect or a difference.

Set up the hypotheses

- ▷ We formally call the claim that we are **hoping to disprove the null hypothesis**. The null hypothesis is generally denoted as H_0 .
- ▷ The test of significance is designed to assess the strength of the evidence against the null hypothesis.
- ▷ Usually the null hypothesis is a statement of **no effect or no difference**.
- ▷ The conclusion we'd like to make is captured in the alternative hypothesis. The alternative hypothesis is generally denoted H_a or H_1 . This is written as the opposite of the null hypothesis and generally suggests that **there is an effect or difference**.

Set up the hypotheses

- ▷ Hypotheses always refer to some population or distribution, **not to a particular sample outcome**. Thus we state H_0 and H_1 in terms of population parameters and not in terms of sample statistics.
- ▷ The alternative hypothesis states that a parameter differs from its null value in a specific direction (an **one-sided alternative**) or in either direction (a **two-sided alternative**).

Set up the hypotheses - one-sided alternative

Example: A gym is interested in whether or not a 6-week weight loss training program they launched has been successful in helping their clients lose weight. To assess this, they took a sample of 30 participants. State the null and alternative hypotheses.

Let's denote the mean weight change, μ , as the population parameter of interest. The gym is specifically interested in whether program participants on average lost weight. Thus they would be interested in an **one-sided alternative** where they seek to claim that program participants **lose weight on average**.

A shorthand notation to capture these in an easy to read form:

$H_0 : \mu = 0$ (there is **no effect on weight change** of participants)

$H_1 : \mu < 0$ (**participants lose weight** on average)

Set up the hypotheses - two-sided alternative

Example: *County officials are interested in measuring a particular chemical in water sources in the county. High levels of this chemical are harmful as are low levels. Either indicate a need for intervention.*

Normal levels of this chemical are **15 parts per million (ppm)**. County officials will start an intervention program if the chemical mean level in the water sources is **different than 15 ppm**.

State the null and alternative hypotheses.

Let's denote the mean level of the chemical, μ , as the population parameter of interest. The alternative hypothesis is that the levels are different than 15 ppm. It is a two-sided alternative.

$H_0 : \mu = 15$ (the mean level of the chemical is normal)

$H_1 : \mu \neq 15$ (the mean level of the chemical is abnormal)

Test Statistics

Significance tests use data from a sample to evaluate the evidence against the null hypothesis.

Significance tests generally compare the value of the population parameter in the null hypothesis to the value of the estimate of the population parameter from the sample.

The **test statistic** is generally a measurement of how far the point estimate is from the expected value under the null.

The test statistic for hypotheses about the mean μ is the z statistic:

$$Z = \frac{\bar{x} - \mu}{\frac{\sigma}{\sqrt{n}}}$$

z-test

A statistical test to determine whether there is a **significant difference between the sample mean and the population mean** when the population variance is known or the sample size is large (typically $n \geq 30$) if **standard normal distribution (Z-distribution)** applies.

n is Large ($n \geq 30$) and σ is unknown

A z-test is any statistical test for which the distribution of the test statistic under the null hypothesis can be approximated by a normal distribution.

Because of the central limit theorem, many test statistics are approximately normally distributed for large samples.

Many statistical tests can be conveniently performed as approximate Z-tests if

- ▷ the sample size is large $n \geq 30$
- ▷ the population variance is known or unknown

Confidence interval is $\bar{x} \pm z_{CL} \times \frac{\sigma}{\sqrt{n}}$ or $\bar{x} \pm z_{CL} \times \frac{S}{\sqrt{n}}$

Note: If the population variance is unknown (and therefore has to be estimated from the sample itself) and **the sample size is not large ($n < 30$)**, the Student's t-test may be more appropriate.

When to Use a Z-Test

a. Known Population Variance

When the population standard deviation (σ) is known.

Example: Quality control in manufacturing where historical data provides a known variance.

b. Large Sample Sizes ($n \geq 30$)

Central Limit Theorem (CLT): For large samples, the sampling distribution of the mean approximates a normal distribution, regardless of the population's distribution.

Implication: Even if σ is unknown, with a large n , the Z-test can be a good approximation.

c. Comparing Sample and Population Means

One-Sample Z-Test: Comparing a single sample mean to a known population mean.

Two-Sample Z-Test: Comparing means from two independent large samples when both population variances are known.

d. Proportion Tests

When Testing Proportions: Assessing whether a sample proportion significantly differs from a known population proportion.

```
Is population variance ( $\sigma$ ) known?  
├ Yes: Use Z-test  
└ No:  
    Is sample size large ( $n \geq 30$ )?  
    ├── Yes: Use Z-test  
    └ No: Use t-test
```

Example Exercise:

- Dataset 1: σ known, $n=40$

Z-test

- Dataset 2: σ unknown, $n=25$

t-test

One of the ways to measure how far the point estimate is from the expected value of the population parameter under the null hypothesis is to **calculate the test statistic's associated p-value.**

The **p-value of the test is the probability**, computed assuming H_0 is true, that the test statistic would take a value as extreme or more extreme than that actually observed.

The p-value is a measure of the strength of the evidence against H_0 .

- ▷ **A small p-value** suggests that the observed result was **unlikely to occur if the null hypothesis is in fact true.**
- ▷ **Large p-values**, on the other hand, **do not give evidence against the null hypothesis.**

Meaning of Large p-value

- ▷ If the resulting p-value is large (meaning that the test statistic is small), then the sample did not give evidence against the null hypothesis.
- ▷ A large p-value only means that the data are inconsistent with the null hypothesis, not that we have clear evidence that the null hypothesis is untrue. Statistical tests are set up to look for evidence against the null, **not to prove that the null hypothesis is untrue.**

Calculating p-value of the test

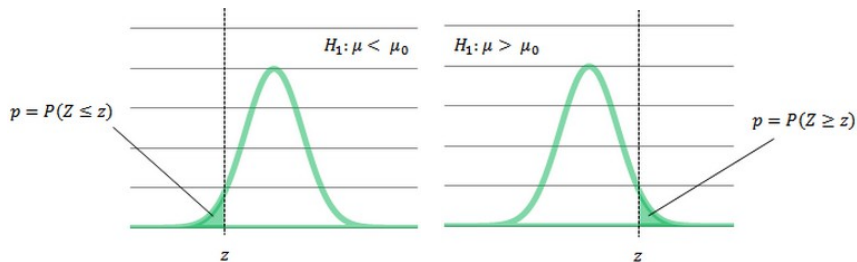
The p-value for the z statistic is calculated using the standard normal distribution.

Depending on the alternative hypothesis, the p-value is calculated as follows:

An one-sided alternative hypothesis.

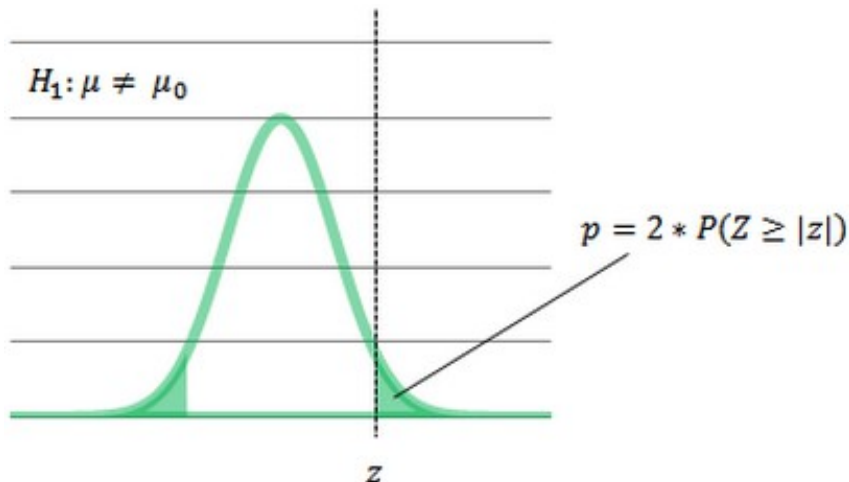
Mean is less than a specified value.

Mean is greater than a specified value.



Calculating p-value of the test - Two Sided

A two-sided alternative hypothesis stating that the population mean is **different** from a specified value.



Example - Calculating P-value for an one-sided test

A gym is interested in whether a 6-week weight loss training program they launched has been successful in helping their clients lose weight. To assess this, they took a sample of 30 participants.

They are interested in testing the following hypotheses:

- ▷ $H_0 : \mu = 0$ (there is no effect on weight change of program participants)
- ▷ $H_1 : \mu < 0$ (program participants lose weight on average)

Suppose we know that for the general population, the standard deviation of changes in weights over a six-week interval is 6 pounds.

The sample mean of the change in weight for the 30 participants in the sample was -2.98 pounds.

Calculate the value of the test statistic and the associated p-value.

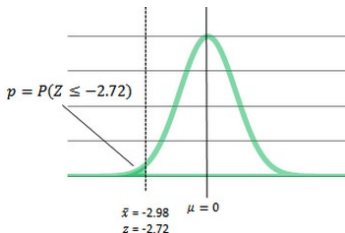
Calculating the value of the test statistic and p-value

$$\sigma = 6, n = 30, \bar{x} = -2.98, \mu_{H_0} = 0$$

The p-value is the probability that the test statistic is -2.72 or more extreme, which is the probability that $Z \leq -2.72$. Using the standard normal table, we can calculate: $P = P(Z \leq -2.72) = 0.0033$

$$z = \frac{\bar{x} - \mu_{H_0}}{\frac{\sigma}{\sqrt{n}}} = \frac{-2.98 - 0}{\frac{6}{\sqrt{30}}} \approx \frac{-2.98}{1.0954} \approx -2.72$$

This is a small p-value. It appears that the sample mean ($= -2.98$) is highly unlikely to have occurred if the true population mean $\mu = 0$. **Thus we have strong evidence against the null hypothesis.**



Example: Chemical in Water - Two-Sided

Normal levels of this chemical are 15 parts per million (ppm).

Samples from 50 water sources throughout the county are taken and the levels of this chemical are measured.

They are interested in testing the following hypotheses:

$H_0 : \mu = 15$ (the mean level of the chemical is normal)

$H_1 : \mu \neq 15$ (the mean level of the chemical is abnormal)

Suppose we know that the population standard deviation is 6.2.

The sample mean from the 50 samples was 16.4 ppm.

Calculate the value of the test statistic and the associated p-value.

Example: Chemical in Water - Two-Sided

Givens are:

$$\bar{x}=16.4, \mu_{H_0}=15, \sigma=6.2, n=50$$

Now, we can just plug these values in to calculate the value of z.

$$z = \frac{\bar{x} - \mu_{H_0}}{\frac{\sigma}{\sqrt{n}}} = \frac{16.4 - 15}{\frac{6.2}{\sqrt{50}}} \approx \frac{1.4}{0.8768} \approx 1.60$$

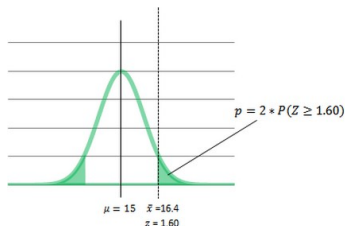
The p-value is the probability that the test statistic is 1.60 or more extreme.

$$H_0 : \mu = 15$$

$$H_1 : \mu \neq 15$$

Example: Chemical in Water - Two-Sided

That is, the p-value is the probability that $Z \geq 1.60$ or $Z \leq -1.60$.



$$\begin{aligned} P &= P(Z \leq -1.60 \text{ or } Z \geq 1.60) = P(Z \geq 1.60) + P(Z \leq -1.60) \\ &= 2 \times P(Z \geq 1.60) = 2 \times 0.0548 = 0.1096 \end{aligned}$$

It appears that the sample mean that we observed ($\bar{x} = 16.4$) is moderately likely to have occurred if the true population mean was 15 ppm (if $\mu = 15$).

We don't have strong evidence against the null hypothesis.

Evaluating Hypotheses using p-value - Significance Level

In order to decide whether to reject the null hypothesis, we can either compare: **the p-value to a pre-defined significance level** the test statistic to a critical value.

For either method, the **significance level** needs to be pre-specified. **It is denoted α (the Greek letter "alpha")**.

The most common choice for **alpha is 0.05** (It is also called standard alpha level). The evidence from the data is so strong that the result obtained would only appear 5% of the time or less if the null hypothesis is in fact true.

- ▷ If the **$p\text{-value} \leq \alpha$** , then we **reject the null hypothesis** in favor of the alternative hypothesis.
- ▷ If the **$p\text{-value} > \alpha$** , then we **fail to reject the null hypothesis**. We do not have sufficient evidence that the null hypothesis is not true.

Evaluating Hypotheses using critical values

The critical value depends on the significance level and whether the test is one- or two-sided.

For a significance test with standard normal distribution ($\mu = 0, \sigma = 1$)

α	10%	5%	2%	1%
1-sided critical value	1.282	1.645	2.054	2.326
2-sided critical value	1.645	1.960	2.326	2.576

If the absolute value of the test statistic $\geq$ the critical value, then we **reject the null hypothesis** in favor of the alternative hypothesis.

If the absolute value of the test statistic $<$ the critical value, then we **fail to reject** the null hypothesis.

Significance Test Result

If the $p\text{-value} \leq \alpha$ or (equivalently) the absolute value of the test statistic $\geq$ the critical value, then we say that the data are "statistically significant at the α level."

In statistics, "Significant" means "not likely to have happened by chance."

The significance level **quantifies** exactly how we are defining "not likely."

Generally, we should report **both the significance level as well as the p-value** as others may be interested in a more or less conservative level of significance.

By provided the exact p-value, the reader may compare the p-value with their own level of significance.

Test of Significance (hypothesis test)

There are 5 key steps in carrying out any significance test:

1. Set up the **hypotheses** and **select the alpha level**
2. Select the appropriate **test statistic**
3. State the **decision rule**
4. **Compute** the test statistic and the associated p-value
5. **State your conclusion**

Example: An One-Sided Test

A gym is interested in whether a 6-week weight loss training program they launched has been successful in helping their clients lose weight. To assess this, they took a sample of 30 participants.

We know that the standard deviation of changes in weights over a six-week interval is 6 pounds.

$$s = 6$$

The sample mean of the change in weight for the 30 participants in the sample was -2.98 pounds.

$$\bar{x} = -2.98, n = 30$$

Perform a significance test to determine whether the weight loss training program has been successful at the $\alpha = 0.05$ level of significance.

An one-sided test example

1. Set up the hypotheses and select the alpha level

$H_0 : \mu = 0$ (there is no effect on weight change of program participants)

$H_1 : \mu < 0$ (program participants lose weight on average)

$\alpha = 0.05$

2. Select the appropriate test statistic

$$Z = \frac{\bar{x} - \mu}{\frac{\sigma}{\sqrt{n}}}$$

3. State the decision rule

Determine the appropriate critical value from the standard normal distribution table associated with a right hand tail probability of $\alpha = 0.05$.

Using the table, the appropriate critical value is 1.645.

Decision Rule: Reject H_0 if $|z| \geq 1.645$

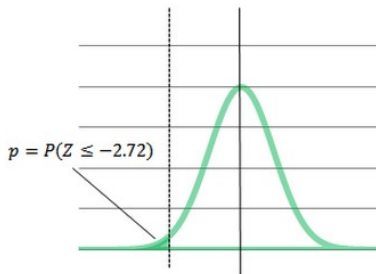
Otherwise, do not reject H_0

An one-sided test example

4. Compute the test statistic and the associated p-value

$$z = \frac{\bar{x} - \mu}{\frac{\sigma}{\sqrt{n}}} = \frac{-2.98 - 0}{\frac{6}{\sqrt{30}}} \approx \frac{-2.98}{1.0954} \approx -2.72$$

$$p = P(Z \leq -2.72) = 0.0033$$



5. Conclusion

Reject H_0 since $|2.72| \geq 1.645$

We have significant evidence at the $\alpha = 0.05$ level that $\mu < 0$.

We reject the null hypothesis that the weight loss program has no effect on weight change of program participants in favor of the alternative hypothesis that program participants lose weight on average ($p = 0.0033$).

A two-sided test example

County officials were interested in measuring a particular chemical in water sources in the county. High levels of this chemical are as harmful as are low levels. Normal levels of this chemical are 15 parts per million (ppm).

- ▷ Samples from 50 water sources throughout the county are taken and the levels of this chemical are measured.
- ▷ Suppose we know that the population standard deviation is 6.2. The sample mean from the 50 samples was 16.4 ppm.

A significance test was conducted to determine whether the mean levels are different than 15 ppm at the $\alpha = 0.05$ level of significance.

However, there was not enough evidence to suggest that the mean levels were not normal.

Given that the significance test did not reject the null hypothesis at the $\alpha = 0.05$ level, would you expect that the 95% confidence interval for the population mean to include 15 ppm or not?

A Two-Sided Test Example

Let's confirm by calculating the confidence interval. To calculate a confidence interval with a confidence level of C for the population mean, μ , we use formula:

$$\bar{x} \pm z \cdot \frac{\sigma}{\sqrt{n}}$$

The 95% confidence interval is calculated:

$$\begin{aligned}\bar{x} \pm z \cdot \frac{\sigma}{\sqrt{n}} &= 16.4 \pm 1.960 \cdot \frac{6.2}{\sqrt{50}} \\ &= 16.4 \pm 1.960 \cdot 0.8768 = 16.4 \pm 1.7186 \\ &\approx (14.68, 18.12)\end{aligned}$$

We are 95% confident that the true mean level is between 14.69 ppm and 18.12 ppm.

The 95% confidence interval contained the null value of 15 ppm since the two-sided significance test at the $\alpha = 0.05$ level did not reject the null hypothesis that $\mu = 15$ (the mean levels are normal).

Significance Tests and Confidence Intervals

There is a relationship between two sided tests conducted with a significance level of α and $1 - \alpha$ confidence intervals.

- ▷ A level α significance test **rejects the null hypothesis** $H_0: \mu = \mu_0$ when the value of μ_0 is not included in the $1 - \alpha$ confidence interval for μ .
- ▷ On the other hand, a level α significance test **fails to reject the null hypothesis** $H_0: \mu = \mu_0$ when the value of μ_0 is included in the $1 - \alpha$ confidence interval for μ .

The conclusion of a significance test (whether or not the null hypothesis is rejected) at the α level of significance can be determined by checking if the null value of the mean as specified by the null hypothesis is contained within the $1 - \alpha$ confidence interval.

More Examples of Hypothesis Testing

Medical Research: Drug Efficacy

A pharmaceutical company claims that their new drug lowers blood pressure by an average of 15 mmHg. A clinical trial is conducted to test this claim.

Hypotheses:

H_0 : The drug lowers blood pressure by ≤ 15 mmHg.

H_1 : The drug lowers blood pressure by > 15 mmHg.

Outcome: After testing, the sample mean reduction is 16 mmHg with a p-value of 0.03.

Decision: Since $0.03 < 0.05$, reject H_0 .

Conclusion: The drug is effective in lowering blood pressure by more than 15 mmHg.

Business: Customer Satisfaction

A company wants to know if their new customer service training increases customer satisfaction scores.

Hypotheses:

H_0 : The training does not increase satisfaction scores ($\mu_{\text{post}} \leq \mu_{\text{pre}}$).

H_1 : The training increases satisfaction scores ($\mu_{\text{post}} > \mu_{\text{pre}}$).

Outcome: Post-training sample mean = 4.5 (on a 5-point scale), pre-training mean = 4.2, p-value = 0.04.

Decision: Since $0.04 < 0.05$, reject H_0 .

Conclusion: The new training program has significantly increased customer satisfaction scores.

Advantages of Hypothesis Testing

Structured Decision-Making: Provides a clear framework for making data-driven decisions.

Objective Evaluation: Reduces personal bias by relying on statistical evidence.

Quantifies Uncertainty: Measures the reliability of conclusions.

Facilitates Comparison: Enables comparison between groups or conditions.

Versatile Applications: Applicable across various fields and scenarios.

Limitations of Hypothesis Testing

Misinterpretation Risks: Common misunderstandings about p-values and confidence levels.

Dependence on Assumptions: Requires data to meet certain statistical assumptions.

Sensitive to Sample Size: Can be affected by too small or too large samples.

Does Not Prove Causation: Indicates association, not causal relationships.

Overreliance on p-Values: May neglect other important metrics like effect size.

Example of Hypothesis Testing, Upsides and Downsides

New Drug Effectiveness

A pharmaceutical company wants to test if a new drug lowers blood pressure more effectively than the current standard drug.

Hypotheses:

H_0 : The new drug is **no more effective** than the standard drug.

H_1 : The new drug is **more effective** than the standard drug.

Result: After conducting a clinical trial with 100 patients, the average blood pressure reduction is significantly higher in the new drug group with a p-value of 0.02.

Upside:

Objective Decision: The low p-value provides strong evidence against H_0 , supporting the claim that the new drug is more effective.

Informed Decision-Making: Helps healthcare providers and policymakers decide whether to adopt the new drug.

Downside:

Assumption Sensitivity: If blood pressure reductions are not normally distributed, the results may be unreliable.

Misinterpretation Risk: Misunderstanding that the p-value indicates the probability of H_0 being true.

Fix it:

Assumption Checking: Use normality tests or non-parametric alternatives if assumptions are violated.

Education on Interpretation: Clarify the correct interpretation of p-values through examples and visual aids.

Common Errors and Solutions

a. Misinterpreting the P-Value: Believing that a p-value of 0.03 means there is a 3% probability that the null hypothesis is true.

- Explain that the p-value represents the probability of obtaining the observed results, or more extreme, assuming the null hypothesis is true.
- Compare it to testing a claim; a low p-value suggests that the claim (null hypothesis) is unlikely to explain the observed data.

b. Choosing the Wrong Hypotheses: Setting up the null and alternative hypotheses incorrectly, such as reversing them.

- H_0 is always the statement of no effect or status quo (current situation before any changes or interventions are made).
- Provide multiple scenarios for students to practice defining H_0 and H_1 .

c. Ignoring Assumptions: Applying hypothesis tests without verifying necessary assumptions (e.g., normality, independence).

- Use plots like histograms or Q-Q plots to assess normality.

d. Calculation Errors

e. Confusing One-Tailed and Two-Tailed Tests

Z test R commands

Z-test (population SD is known = popsd)

```
z <- (mean(data$variable) - mu) / (popsd / sqrt(nrow(data)))

# Then pnorm function can be used to calculate the probability
pnorm(z) or 1 - pnorm(z) or 2 * (1 - pnorm(abs(z)))

# Confidence Interval can be computed by calculating lower and upper
  bounds
lower <- mean(data$variable) - z * popsd / sqrt(nrow(data))
upper <- mean(data$variable) + z * popsd / sqrt(nrow(data))
```

Using "asbio" package

```
install.packages("asbio")

library(asbio)

one.sample.z(null.mu=[mean under null], xbar=mean(data$variable),
sigma=popsd, n=nrow(data), alternative=[alternative], conf=[confidence
  level])

# [alternative] = less , greater , or two.sided
```

MET CS 555 – Foundation of Machine Learning

Module - 2

Lecture - 4: t-test, Scatter Plots, Correlation, Linear Regression

t-test

Test of Significance (hypothesis test)

A statistical method used to determine whether there is a significant difference between the means of two groups, which may be related in certain features. It helps in assessing hypotheses about population means when the sample size is small and the population standard deviation is unknown.

There are 5 key steps in carrying out any significance test:

1. Set up the **hypotheses** and **select the alpha level**
2. Select the appropriate **test statistic**
3. State the **decision rule**
4. **Compute** the test statistic and the associated p-value
5. **State your conclusion**

z-test

n is Large ($n \geq 30$) and σ is unknown

Because of the central limit theorem, many test statistics are approximately normally distributed for large samples.

Many statistical tests can be conveniently performed as approximate Z-tests if

- ▷ the sample size is large $n \geq 30$
- ▷ the population variance is known or unknown

Confidence interval is:

$$\bar{x} \pm Z_{CL} \cdot \frac{\sigma}{\sqrt{n}} \text{ or } \bar{x} \pm Z_{CL} \cdot \frac{S}{\sqrt{n}}$$

If the population variance is unknown (and therefore has to be estimated from the sample itself) and **the sample size is not large ($n < 30$)**, the Student's t-test may be more appropriate.

n is small (< 30) and σ is unknown

When n is not large (< 30), the **Central Limit Theorem does not guarantee** that the distribution of the sample mean is perfectly normal.

When n is not large (< 30) and the standard deviation of the population is not known, the test statistic for hypotheses about the mean μ is the t statistic:

$$t = \frac{\bar{x} - \mu}{\frac{s}{\sqrt{n}}}$$

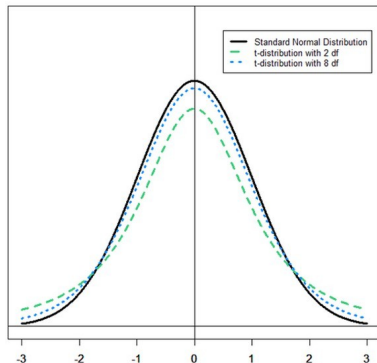
This test is known as the "One-Sample t-test".

t-statistic, t-distribution

t-statistic is not normally distributed.

It is a t-distribution with $n - 1$ degrees of freedom (df)

- ▷ The shape of the distribution varies with the sample size
- ▷ The variability of the t-distributions is slightly greater than that of the standard normal distribution
- ▷ As n increases, s is a better and better estimate of σ , and the variability decreases.



Try the example at: <http://rpsychologist.com/d3/tdist/>

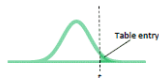


Table entry for p and C is the critical value t with probability p lying to its right and probability C lying between $-t$ and t

Confidence interval in t-statistic

To calculate confidence interval with a confidence level of C for the population mean, μ :

$$\bar{x} \pm t \cdot \frac{s}{\sqrt{n}}$$

t is the appropriate critical value corresponding to the confidence level C with $df = n - 1$.

Also sometimes written as (SE denotes the standard error):

$$\bar{x} \pm t \cdot SE$$

- ▷ t-statistic may also be used in cases where $n \geq 30$ but the population distribution is not normally distributed.
- ▷ When the population is not perfectly normally distributed, the t-distribution provides an adequate approximation.

t-statistic: An Example

Biologists studying the healing skin wounds measured by the rate at which new cells closed a razor cut made in the skin of a newt.



The data from **18 newts** were measured. The **sample mean and standard deviation** of the healing rates were **25.67 and 8.345** micrometers per hour. Calculate the **95% confidence interval** for the mean healing rate in the population of all newts.

t-statistic: An Example

Biologists studying the healing skin wounds measured by the rate at which new cells closed a razor cut made in the skin of a **newt**.



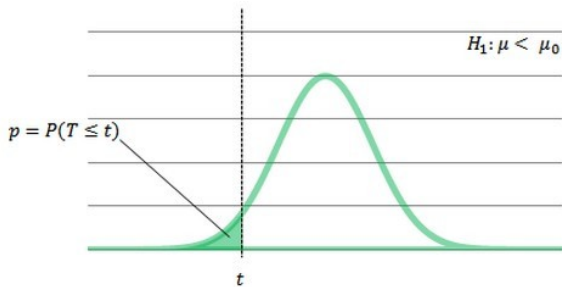
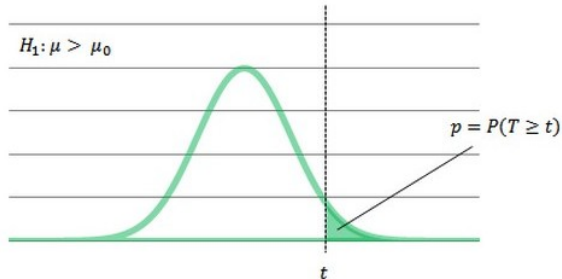
The data from **18 newts** were measured. The **sample mean and standard deviation** of the healing rates were **25.67 and 8.345** micrometers per hour. Calculate the **95% confidence interval** for the mean healing rate in the population of all newts.

$df = n - 1 = 17$, t critical value is 2.11

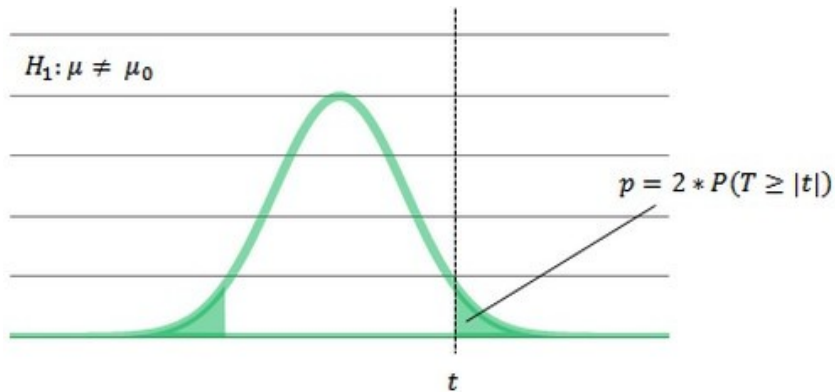
$$\begin{aligned}\bar{x} \pm t \cdot \frac{s}{\sqrt{n}} &= 25.67 \pm 2.11 \cdot \frac{8.345}{\sqrt{18}} = 25.67 \pm 2.11 \cdot 1.9669 \\ &\approx 25.67 \pm 4.15 \approx (21.52, 29.82)\end{aligned}$$

We are 95% confident that the true mean healing rate for all newts is between 21.52 and 29.82 micrometers per hour.

P value calculation in t-statistic - One-Sided



P value calculation in t-statistic - Two-Sided



t-statistic - a one-sided example

A scientist wishes to test the claim that great white sharks average 20 feet in length.

To test this, he measures 10 great white sharks.
The measurements are 18.1, 23.4, 23.8, 24.1,
22.5, 19, 25.4, 23.1, 16.5, 26.7



Are these data good evidence that great white sharks are longer than 20 feet in length at the $\alpha = 0.10$ level of significance?

t-statistic - a one-sided example

1. Set up the hypotheses and select the alpha level

$H_0 : \mu = 20$ (the mean length of great white sharks is 20 ft)

$H_1 : \mu > 20$ (the mean length of great white sharks is greater than 20 ft)

$\alpha = 0.10$

2. Select the appropriate test statistic

$$t = \frac{\bar{x} - \mu}{\frac{s}{\sqrt{n}}}$$

3. State the decision rule

Determine the appropriate critical value from the standard t-distribution associated with a right hand tail probability of $\alpha = 0.10$ based on $df = 9$.
Critical value is 1.383

**Decision Rule: Reject H_0 if $t \geq 1.383$
Otherwise, do not reject H_0**

t-statistic - a one-sided example

4. Compute the test statistic and the associated p-value

$$t = \frac{\bar{x} - \mu}{\frac{s}{\sqrt{n}}} = \frac{22.27 - 20}{\frac{3.31}{\sqrt{10}}} \approx \frac{2.27}{1.05} \approx 2.16$$

5. Conclusion

Reject H_0 since 2.16 is greater than 1.383

We have significant evidence at the $\alpha = 0.10$ level that the mean length of great white sharks is longer than 20 feet

The sample mean was 22.27 feet.

We reject the null hypothesis that the mean length of great white sharks is 20 feet. P-value associated with the t-statistic of 2.16 is between 0.05 and 0.025

P-value associated with the t-statistic

P-value associated with the t-statistic of 2.16 is between 0.05 and 0.025

df	Upper Tail Probability, p											
	0.25	0.2	0.15	0.1	0.05	0.025	0.02	0.01	0.005	0.0025	0.001	0.0005
1	1.000	1.376	1.963	3.078	6.314	17.06	15.895	31.821	63.657	127.321	318.309	636.619
2	0.816	1.061	1.386	1.886	2.920	4.303	4.849	6.965	9.925	14.069	22.327	31.599
3	0.765	0.978	1.250	1.638	2.353	3.182	3.482	4.541	5.841	7.453	10.215	12.924
4	0.741	0.941	1.190	1.533	2.132	2.776	2.999	3.747	4.604	5.598	7.173	8.610
5	0.727	0.920	1.156	1.476	2.015	2.571	2.757	3.365	4.032	4.773	5.893	6.869
6	0.718	0.906	1.134	1.440	1.943	2.447	2.612	3.143	3.707	4.317	5.208	5.959
7	0.711	0.896	1.119	1.415	1.895	2.365	2.517	2.998	3.499	4.029	4.785	5.408
8	0.706	0.889	1.108	1.397	1.860	2.306	2.449	2.896	3.355	3.833	4.501	5.041
9	0.703	0.883	1.100	1.383	1.833	2.262	2.398	2.821	3.250	3.690	4.297	4.781
10	0.700	0.879	1.093	1.372	1.812	2.228	2.359	2.764	3.169	3.581	4.144	4.587

t test R commands

```
# Calculating probabilities from t-statistics
> p <- pt([t statistics], df = [degree of freedom])

# It calculates the area to the left of a given t-statistics
# Calculating t-statistics from probabilities
> t <- qt([probability], df = [degree of freedom])

# It find the t-statistic with the specified area to the left

# t.test function
> t.test(data$variable, mu=[mean under null], alternative=[alternative],
         conf.level=[confidence level])

# [alternative] = "less", "greater", or "two.sided"
```

NOTE: must use "two.sided" option if you want to calculate confidence intervals. Confidence intervals from other options are not correct.

t test R commands

```
> pt(2.76, df=4)
# [1] 0.9745752

> qt(0.975, df=4)
# [1] 2.776445

> qt(0.025, df=4)
# [1] -2.776445
```

Table B. *t*-Distribution Critical Values

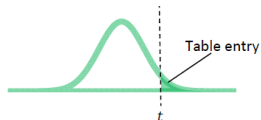


Table entry for p and C is the critical value t with probability p lying to its right and probability C lying between $-t$ and t

df	Upper Tail Probability, p											
	0.25	0.2	0.15	0.1	0.05	0.025	0.02	0.01	0.005	0.0025	0.001	0.0005
1	1.000	1.376	1.963	3.078	6.314	12.706	15.895	31.821	63.657	127.321	318.309	636.619
2	0.816	1.061	1.386	1.886	2.920	4.303	4.849	6.965	9.925	14.089	22.327	31.599
3	0.765	0.978	1.250	1.638	2.353	3.182	3.482	4.541	5.841	7.453	10.215	12.924
4	0.741	0.941	1.190	1.533	2.132	2.776	2.999	3.747	4.604	5.598	7.173	8.610

R Commands: t-test

A scientist wishes to test the claim that great white sharks average 20 feet in length.

The measurements are 18.1, 23.4, 23.8, 24.1, 22.5, 19, 25.4, 23.1, 16.5, 26.7.

Are these data good evidence that great white sharks are longer than 20 feet in length at the $\alpha = 0.10$ level of significance?

```
> shark_len = c(18.1, 23.4, 23.8, 24.1, 22.5, 19, 25.4, 23.1, 16.5, 26.7)
> t <- (mean(shark_len) - 20) / (sd(shark_len)/sqrt(length(shark_len)))
> t
[1] 2.160244

> p <- 1 - pt(t, df=length(shark_len)-1)
> p
[1] 0.02952202

> t.test(shark_len, mu=20, alternative="greater", conf.level=0.9)
```

t-statistic: a two-sided example

Biologists studying the healing skin wounds measured by the rate at which new cells closed a razor cut made in the skin of a newt.



In another experiment, the researchers used the same 18 newts and measured the healing rate after applying a topical therapy that was developed to help speed up healing rates.

The difference between the healing rate for each newt with and without the topical treatment were calculated where positive numbers indicated an increase in the healing rate.

The resulting mean of the differences in the sample was 3.2 and the standard deviation of the differences in the sample was 4.5.

Test whether or not the topical treatment changed the healing rate of the newts at the $\alpha = 0.10$ level of significance.

t-statistic: a two-sided example

1. Set up the hypotheses and select the alpha level

$H_0: \mu = 0$ (the mean difference is 0, the treatment **has no effect** on healing)

$H_1: \mu \neq 0$ (the difference is not 0, the treatment has **an effect** on healing)

$\alpha = 0.10$

2. Select the appropriate test statistic

Since n is small and the sd of the population is unknown

$$t = \frac{\bar{x} - \mu}{\frac{s}{\sqrt{n}}}$$

3. State the decision rule

Critical value from the standard t-distribution associated with a right hand tail probability of $\frac{\alpha}{2} = \frac{0.1}{2} = 0.05$ based on $df = 17$ is

1.74.
Decision Rule: Reject H_0 if $|t| \geq 1.74$, Otherwise, do not reject H_0

t-statistic: a two-sided example

4. Compute the test statistic and the associated p-value

$$t = \frac{\bar{x} - \mu}{\frac{s}{\sqrt{n}}} = \frac{3.2 - 0}{\frac{4.5}{\sqrt{18}}} \approx \frac{3.2}{1.06} \approx 3.02$$

5. Conclusion

Reject H_0 since 3.02 is greater than 1.74

We have significant evidence at the $\alpha = 0.05$ level that the healing rate is faster with the topical cream than without.

The topical cream increased the healing rate by 3.2 micrometers per hour. We reject the null hypothesis that the mean topical cream had no effect on healing rate of the newts.

Calculate p-value.

t-statistic: a two-sided example - Calculate p-value

The t-statistic lies between the critical values for 0.005 and 0.0025, the p-value associated with the t-statistic of 3.02 is between 2×0.005 and 2×0.0025 (since it is a **two-sided test**).

The p-value is between 0.005 and 0.010.

P value is equal to 0.008 (which indeed is between 0.005 and 0.010).

df	Upper Tail Probability, p											
	0.25	0.2	0.15	0.1	0.05	0.025	0.02	0.01	0.005	0.0025	0.001	0.0005
1	1.000	1.376	1.963	3.078	6.314	12.706	15.895	31.821	63.657	127.321	318.309	636.619
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5	0.727	0.920	1.156	1.476	2.015	2.571	2.757	3.365	4.032	4.773	5.893	6.869
6	0.718	0.906	1.134	1.440	1.943	2.447	2.612	3.143	3.707	4.317	5.208	5.959
7	0.711	0.896	1.119	1.415	1.895	2.365	2.517	2.998	3.499	4.029	4.785	5.408
8	0.706	0.889	1.108	1.397	1.860	2.306	2.449	2.896	3.355	3.833	4.501	5.041
9	0.703	0.883	1.100	1.383	1.833	2.262	2.398	2.821	3.250	3.690	4.297	4.781
10	0.700	0.879	1.093	1.372	1.812	2.228	2.359	2.764	3.159	3.581	4.144	4.587
11	0.697	0.876	1.088	1.363	1.796	2.201	2.328	2.718	3.106	3.497	4.025	4.437
12	0.695	0.873	1.083	1.356	1.782	2.179	2.303	2.681	3.055	3.428	3.930	4.318
13	0.694	0.870	1.079	1.350	1.771	2.160	2.282	2.650	3.012	3.372	3.852	4.221
14	0.692	0.868	1.076	1.345	1.761	2.145	2.264	2.624	2.977	3.326	3.787	4.140
15	0.691	0.866	1.074	1.341	1.753	2.131	2.249	2.602	2.947	3.296	3.733	4.073
16	0.690	0.865	1.071	1.337	1.746	2.120	2.235	2.583	2.924	3.267	3.686	4.015
17	0.689	0.863	1.069	1.333	1.740	2.110	2.224	2.567	2.898	3.222	3.646	3.965
18	0.688	0.862	1.067	1.330	1.734	2.101	2.214	2.552	2.878	3.197	3.610	3.922
19	0.688	0.861	1.066	1.328	1.729	2.093	2.205	2.539	2.861	3.174	3.579	3.883
20	0.687	0.860	1.064	1.325	1.725	2.086	2.197	2.528	2.845	3.153	3.552	3.850

About the above example - Matched Pairs Design

- ▷ Studies where the same individual or experimental unit is given more than one treatment and the variable of interest is the change between the treatments is called a **matched pairs design**.
- ▷ We applied **one-sample** test procedures to analyze the observed differences.
 1. either the one sample z-test or
 2. the one sample t-test
- ▷ Since **two measurements are taken on the same observational unit** in this type of design, the observations must first be compared and the analysis needs to take place on the differences.

About the above example - Matched Pairs Design

- ▷ Next, we will discuss tests for two samples (two-sample test).
Note that the two-sample tests do not apply to this situation.

When t-test is not applicable ...

- ▷ **Normal Distribution.** In real-world applications, data is rarely perfectly normally distributed. The validity of the procedures involving the t-distribution rely on how different the distribution of the data is from the normal distribution.
- ▷ **Outliers.** The presence of outliers or strong skewness in the distribution is indicative of severe departure from this assumption. As the **mean and the standard deviation are sensitive to outliers and strong skewness**, the procedures based on the t-distribution where the **sample sizes are small** are also greatly influenced by such departures from normality.
- ▷ **Skewness.** It is essential to check the distribution of your sample data and check for strong skewness and outliers before you use t-procedures. If strong skewness exists or there are outliers present, use of the t-procedures **is not recommended** (especially when using these procedures when **$n < 30$**).

Two-sample problems

Two-sample problems

- ▷ We are interested in comparing the distribution of the **sample means from two populations**.
- ▷ We have separate samples from each population and we want to make conclusions about the characteristics of the **two samples for a particular parameter**.
- ▷ In the two sample setting, one may be interested in **comparing the centers (the means) or the spreads (the standard deviations) between the two populations**.
- ▷ Two-sample tests for means are one of **the most common situations encountered in statistical practice**. It is the focus of our course.

Conditions to use two-sample procedures

In the two-sample setting, conditions necessary for inference include:

1. **Independence.** Samples must be independent (not influencing each other) and randomly selected from the two distinct populations of interest
2. **Same Measurement.** The variable of interest must be measured in the same way in each of the populations.
3. **Normal Distribution.** The parameter of interest should be normally distributed (or at least have similar shapes and without outliers)

The notation used for two-sample problems

We are interested in estimating $\mu_1 - \mu_2$ or testing the null hypothesis of no difference between means, $\mu_1 - \mu_2 = 0$ or, $\mu_1 = \mu_2$.

- ▷ $\bar{x}_1$ and $\bar{x}_2$ estimate μ_1 and μ_2
- ▷ s_1 and s_2 estimate σ_1 and σ_2 .

Population	Variable	Population		Sample		
		Mean	Standard Deviation	Sample size	Mean	Standard Deviation
1	x_1	μ_1	σ_1	n_1	$\bar{x}_1$	s_1
2	x_2	μ_2	σ_2	n_2	$\bar{x}_2$	s_2

Two-sample Test of Means

The standard deviation of the **difference** in the sample means, $\bar{x}_1 - \bar{x}_2$, is :

$$\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}$$

Given that σ_1 and σ_2 are generally unknown, we estimate by using the sample standard deviations:

$$\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$$

The above quantity is referred to as the **Standard Error of the difference in sample means**.

Two-sample Test of Means

The two sample t-statistic used for hypothesis testing is calculated by:

$$t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

This represents how far the difference in sample means is from 0 in standard deviations units.

Two-sample Test of Means

The formula is sometimes written as:

$$t = \frac{(\bar{x}_1 - \bar{x}_2) - (\mu_1 - \mu_2)}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

This is for tests of the null hypothesis: $\mu_1 - \mu_2 = d$

Since generally we are concerned with testing if the means in the population are different, d is set to 0 which leaves us with the formula presented above.

Two-sample Test: degree of freedom of t-distribution

The t-statistic approximately has a t-distribution.

Its degrees of freedom has a very complicated formula:

$$df = \frac{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2} \right)^2}{\frac{\left(\frac{s_1^2}{n_1} \right)^2}{n_1 - 1} + \frac{\left(\frac{s_2^2}{n_2} \right)^2}{n_2 - 1}}$$

Without any software, a conservative approach would be to estimate the degrees of freedom by taking **the minimum of the quantities $n_1 - 1$ and $n_2 - 1$** .

- ▷ This approach will always result in a **p-value that is slightly larger** than the actual value.
- ▷ Using this approach you will not be far off unless the sample sizes are both quite small and are not equal to each other.
- ▷ This approximation becomes better and better as the sample sizes in each of the populations increase.

Two-sample Test: An Example - Polyester Decay

How quickly do synthetic fabrics such as polyester decay in landfills?

In order to assess how quickly polyester decays over time in landfills, a researcher buried strips of the material in the soil for different lengths of time and then tested the force required to break them (as a measure of decay).



Two-sample Test: an example

Here is the data collected:

Group	Sample Data				
2 weeks	118	126	126	120	129
16 weeks	124	98	110	140	110

Lower breaking strength is indicative of decay.

Test whether or not the breaking strengths of polyester strips buried for 2 weeks is greater than the breaking strengths of those buried for 16 weeks.

Perform the test at the $\alpha = 0.10$ level of significance.

Example: Two-Sample Test - Hypotheses Test

1. Set up the hypotheses and select the alpha level

$H_0 : \mu_1 = \mu_2$ (the mean breaking strengths are the same)

$H_1 : \mu_1 > \mu_2$ (the mean breaking strengths are greater after 2 weeks versus 16 weeks)

$\alpha = 0.10$

2. Select the appropriate test statistic

$$t = \frac{(\bar{x}_1 - \bar{x}_2) - (\mu_1 - \mu_2)}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

since n is small and the sd of the population is unknown

Example: Two-Sample Test - Hypotheses Test

3. State the decision rule

- ▷ Determine the appropriate critical value from the standard t-distribution table associated with a right hand tail probability of $\alpha = 0.10$.
- ▷ We would choose $df = 5 - 1 = 4$
- ▷ Using the table or software, the appropriate critical value is 1.533
Decision Rule: Reject H_0 if $t \geq 1.533$
Otherwise, do not reject H_0

Example: Two-Sample Test - Hypotheses Test

4. Compute the test statistic and the associated p-value

$$t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}} = \frac{123.8 - 116.4}{\sqrt{\frac{4.60^2}{5} + \frac{16.09^2}{5}}} = \frac{7.4}{7.484} = 0.9889$$

5. Conclusion

- ▷ Fail to reject H_0 since 0.9889 is less than 1.53
- ▷ **We do not have significant evidence at the $\alpha = 0.10$ level to conclude that polyester decays more in 16 weeks than in 2 weeks**

•

Two-Sample Confidence Interval for Comparison of Means

The confidence interval for the difference in population means is:

$$(\bar{x}_1 - \bar{x}_2) \pm t \cdot \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$$

Confidence intervals are typically of the form:

$$\text{Estimate} \pm \text{critical value} \cdot SE_{\text{estimate}}$$

Two-sample Test: Calculating confidence interval

Let's return to the example on breaking strengths of polyester.

Calculate the 90% confidence interval for the difference in breaking strengths between groups.

Group	Sample Data				
2 weeks	118	126	126	120	129
16 weeks	124	98	110	140	110

Two-sample Test: Calculating confidence interval

$$(\bar{x}_1 - \bar{x}_2) \pm t \cdot \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$$

90% confidence interval can be estimated from the standard t-distribution table associated with a right hand tail probability of $\frac{\alpha}{2} = \frac{0.10}{2} = 0.05$

Two-sample Test: Calculating confidence interval

$$(\bar{x}_1 - \bar{x}_2) \pm t \cdot \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$$

90% confidence interval can be estimated from the standard t-distribution table associated with a right hand tail probability of $\frac{\alpha}{2} = \frac{0.10}{2} = 0.05$

- ▷ A conservative approach would be to estimate the degrees of freedom by taking the minimum of the quantities $n_1 - 1$ and $n_2 - 1$.
- ▷ We would choose $df = 5 - 1 = 4$
- ▷ **Critical value is 2.13**

Two-sample Test: Calculating confidence interval

$$(\bar{x}_1 - \bar{x}_2) \pm t \cdot \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$$

90% confidence interval can be estimated from the standard t-distribution table associated with a right hand tail probability of $\frac{\alpha}{2} = \frac{0.10}{2} = 0.05$

- ▷ A conservative approach would be to estimate the degrees of freedom by taking the minimum of the quantities $n_1 - 1$ and $n_2 - 1$.
- ▷ We would choose $df = 5 - 1 = 4$
- ▷ **Critical value is 2.13**

$$(123.8 - 116.4) \pm 2.132 \cdot \sqrt{\frac{4.60^2}{5} + \frac{16.09^2}{5}} = 7.4 \pm 2.132 \cdot 7.484$$
$$\approx -8.56 \text{ to } 23.36$$

We are 90% confident that the mean difference in breaking strengths is between -8.56 and 23.36.

Pooled Two-Sample t-Test

- ▷ The approach presented for two-sample tests work regardless of whether or not the sample **standard deviations are the same**.
- ▷ When equal variances between groups are assumed, another methodology, called the **pooled two-sample t-test** (or the two-sample tests for equal variances), can be used.
However, the presented methodology is preferred in all cases given its generality and lack of this assumption.
- ▷ The pooled two-sample t-test is **more sensitive to departures from normality**.
- ▷ In cases that we have **same sample sizes**, the pooled two-sample t-test has the same results as the methodology presented above.

R commands – Two-sample t test

```
# t.test function  
> t.test(x, y, alternative=[alternative], conf.level=[confidence level]  
  
# [alternative] = less , greater , or two.sided  
# var.equal =FALSE (default) should always be used
```

Be careful: for one-sided tests, order that you put x and y matters!

NOTE: must use **"two.sided"** option if you want to calculate confidence intervals. Confidence intervals from other options are not correct.

Two-sample Test: An Example - Polyester Decay

```
> decay = read.csv("decay.csv")
> summary(decay$strength[decay$weeks==2])
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
 118.0  120.0   126.0   123.8  126.0   129.0
> summary(decay$strength[decay$weeks==16])
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
  98.0  110.0   110.0   116.4  124.0   140.0
```

1. Set up the hypotheses and select the alpha level

$H_0 : \mu_1 = \mu_2$ (the mean breaking strengths are the same)

$H_1 : \mu_1 > \mu_2$ (the mean breaking strengths are greater after 2 weeks versus 16 weeks)

$\alpha = 0.10$

2. Select the appropriate test statistic

Use t-test as n is small and population sd is unknown

Two-sample Test: An Example - Polyester Decay

3. State the decision rule

Decision Rule: Reject H_0 if $p\text{-value} \leq \alpha$

Otherwise, do not reject H_0

4. Compute the test t statistic and the associated p-value

```
> t.test(decay$strength[decay$weeks==2], decay$strength[decay$weeks==16], alternative="greater", conf.level=0.9)
```

```
t = 0.9889, df = 4.651, p-value = 0.1857
```

5. Conclusion

- ▷ Fail to reject H_0 since p-value is greater than α
- ▷ **We do not have significant evidence at the $\alpha = 0.10$ level to conclude that polyester decays more in 16 weeks than in 2 weeks.**

Two-sample Test Confidence Interval : An Example - Polyester Decay

```
> t.test(decay$strength[decay$weeks==2], decay$strength[decay$weeks  
==16], alternative="two.sided", conf.level=0.95)
```

```
t = 0.9889, df = 4.651, p-value = 0.3713
```

```
alternative hypothesis: true difference in means is not equal to 0
```

```
95 percent confidence interval:
```

```
-12.2789 27.0789
```

Closing Remarks

- ▷ two-sample tests described is quite **robust to departures from normality assumption.**
- ▷ This is particularly true **when the sample sizes n_1 and n_2 are equal.**
- ▷ As long as the underlying distributions have **similar shapes and the sample sizes are both ≥ 5**

Closing Remarks

- ▷ **Larger sample sizes are needed when the distributions have different shape.**
- ▷ **Lack of sensitivity to skewness and departures of normality make it so widely used in statistics.**
- ▷ **z-tests** can be used in the setting where n_1 and n_2 are large (both ≥ 30).
In practice, they are rarely used given the robustness of the t-tests to non-normality.

R Commands for Numerical Summaries



R commands: numerical summaries for two-sample tests

```
# Use aggregate or tapply function to summarize data by population
> aggregate(data$variable, by=list(data$population), FUN=function)

> tapply(data$variable, data$group, mean)
```

Can be used for any function (summary, sd)

R commands: numerical summaries for two-sample tests

Side by side box plots

```
> boxplot(data$variable ~ data$group)
```

Bar graph with confidence intervals

```
> install.packages("gplots")
```

```
> attach(data)
```

```
> means <- tapply(variable, group, mean)
```

```
> lower <- tapply(variable, group, function(v) t.test(v)$conf.int[1])
```

```
> upper <- tapply(variable, group, function(v) t.test(v)$conf.int[2])
```

```
> barplot2(means, plot.ci=TRUE, ci.l=lower, ci.u=upper, names.arg=[  
  labels])
```

R Commands: numerical summaries for two-sample tests

Side by side box plots

```
> boxplot(data$variable~data$group)
```

Bar graph with confidence intervals

```
> install.packages("gplots")
```

```
> attach(data)
```

```
> means <- tapply(variable, group, mean)
```

```
> lower <- tapply(variable, group, function(v) t.test(v)$conf.int[1])
```

```
> upper <- tapply(variable, group, function(v) t.test(v)$conf.int[2])
```

```
> barplot2(means, plot.ci=TRUE, ci.l=lower, ci.u=upper, names.arg=[  
  labels])
```

R Commands: numerical summaries for two-sample tests

```
> attach(decay)

# Side by side box plots
> boxplot(strength ~ weeks)

# Bar graph with confidence intervals
> means <- tapply(strength, weeks, mean)

# Using t.test function to find out upper and lower bounds
> lower <- tapply(strength, weeks, function(v) t.test(v)$conf.int[1])
> upper <- tapply(strength, weeks, function(v) t.test(v)$conf.int[2])

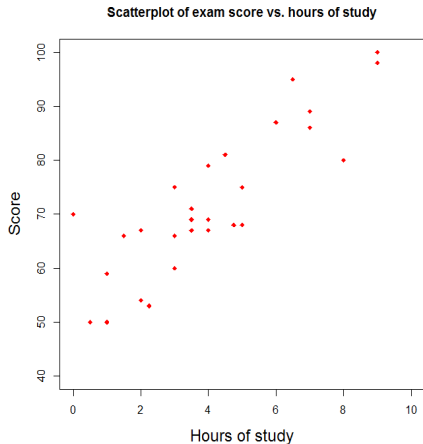
# put the boundaries on top of a bar plot
> barplot2(means, plot.ci=TRUE, ci.l=lower, ci.u=upper, names.arg=c(      2
  weeks ,      16 weeks ))

# a horizontal line
> abline(h=0)
```

Scatterplots

Scatterplots

- Display the relationship between two **continuous or quantitative factors**.
- Shows the relationship between **two paired factors**
- **Each "pair" of data** is shown with one single point.



An example - Scatterplots

Is there an association between the number of hours spent studying and the performance on the final exam?

Use `plot()` function to draw the scatterplot

```
> plot(data$explanatoryvariable, data$responsevariable)
```

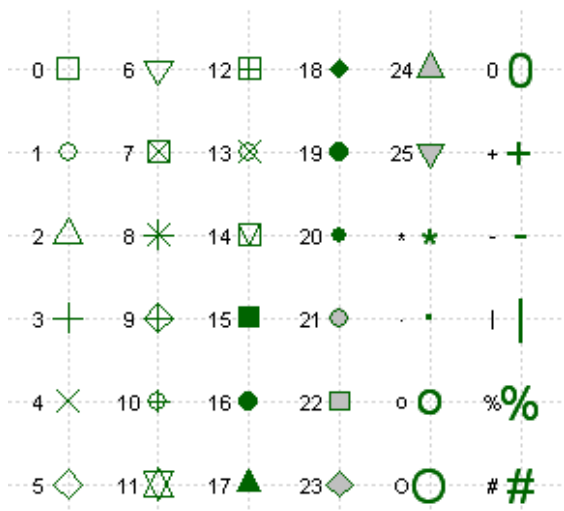
Many Attributes:

- ▷ Use **`main`, `xlab`, and `ylab`** to label the picture appropriately
- ▷ Use **`xlim` and `ylim`** to control x and y axes
- ▷ Change the type of point using **`pch`** and/or the color of the point using **`col`**
- ▷ Change the size of the points or the labels using `cex`, `cex.axis`, `cex.main`, etc.

See **`?plot()`** for documentation

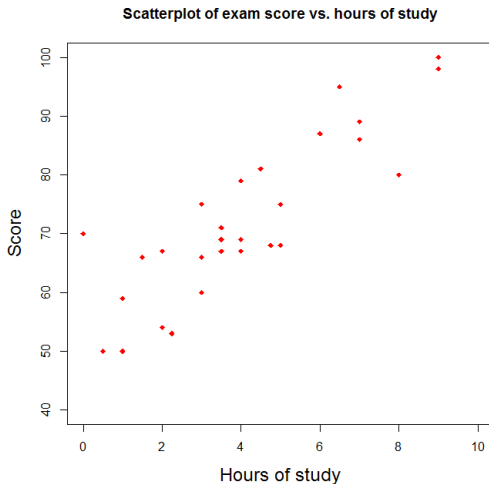
Plot() options for pch

plot symbols : pch =



Scatterplot Example

```
> attach(student)
> plot(study.hours, score, main="Scatterplot of exam score vs. hours of study", xlab="Hours of study", ylab="Score", xlim=c(0,10), ylim=c(40,100), pch=18, col="red", cex.lab=1.5)
```



Response versus Explanatory Variables

- ▷ Generally, we put
 - ▷ the response variable (the outcome of interest) on the **y-axis** (vertical axis) and
 - ▷ the explanatory variable on the **x-axis**.
- ▷ **Explanatory variables** are also called **independent variables**.
- ▷ **Response variables** are called **dependent variables** (due to the fact that the response variable may depend on the explanatory variable).

Response versus Explanatory Variables

- ▷ If there is a **temporal relationship** (if one variable comes before another) then the **explanatory variable is the generally the one that occurs first.**

1. Study Hours → Exam Score

Explanation: Study hours occur before exam scores. Students study for a certain number of hours **before** taking the exam.

2. Advertising Spend → Sales Revenue

Explanation: Advertising spend occurs before the sales revenue is realized. Increased advertising is intended to boost sales

- ▷ If there is **not a temporal relationship**, then to figure out which may be a better choice for the explanatory variable, you may ask yourself **which factor may** depend on the other.

1. Height Shoe Size or Height Academic Performance

Explanation: No clear time order; height influences shoe size. Height and shoe size are measured at the same time; one does not occur **before** the other.

2. Caloric Intake Body Weight

Explanation: No Clear Temporal Order: Both diet and body weight can be measured at the same time. One does not happen **before** the other.

Example - Investing in an irrigation system

A farm in upstate New York is evaluating whether or not to invest in an irrigation system. They have data over the last few years on average rain fall (in inches during the growing season) and pounds of apples produced. If the amount of rain fall is associated with the crop yield, they may choose to make the investment.

Which is the response variable and which is the explanatory variable in this case?

Rain fall during the growing season occurs temporally **before the crops produce their yield.**

A choice is that the response variable is pounds of apples and the explanatory variable is average rain fall.

Example - Are SAT math and verbal scores associated?

Are SAT math and verbal scores associated? To find out, a high school collects the math and verbal scores of their students.

If they were to make a scatterplot of these data, which should go on the xaxis?

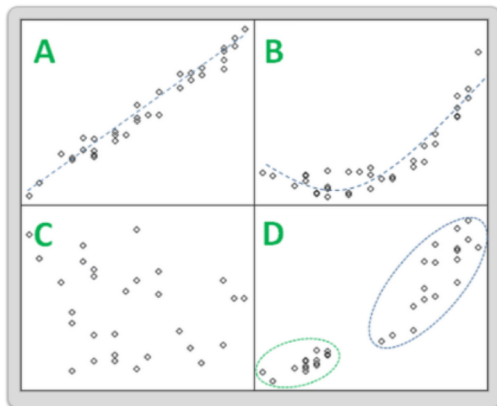
Generally the explanatory variable goes on the x axis of the scatterplot. However, in this case, **there is not a clear choice for the response and explanatory variables.**

There is no reason to think that ones SAT math score depends on ones SAT verbal score, or vice versa. As such, either score could go on the x axis.

Interpreting Scatterplots - Form

Relationships between variables:

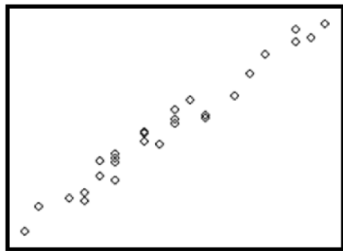
- ▷ **Linear** (where the points tend towards a straight line pattern)
- ▷ **Curved** (where the points tend toward a U-shape or arced pattern)
- ▷ **Random** (where the points don't seem to follow any pattern)
- ▷ **Clusters** may also be apparent



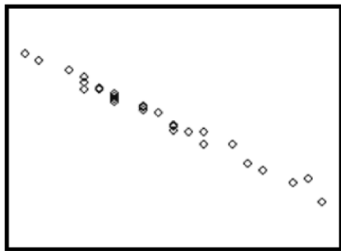
Interpreting scatterplots - Direction

The relationship between two factors is:

- ▷ **"Positively Associated"** when as one factor increases in value the other factor also tends to increase in value
- ▷ **"Negatively Associated"** when as one factor increases in value the other factor tends to decrease in value

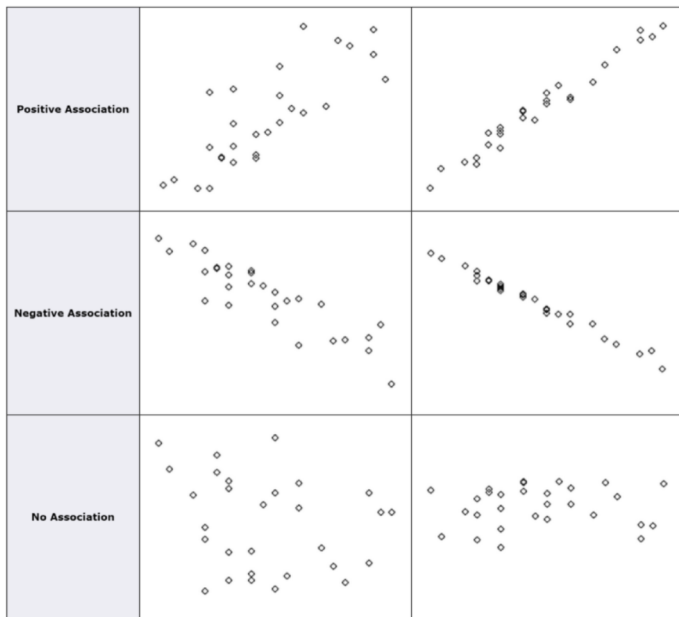


Positive Association



Negative Association

Interpreting scatterplots - Direction

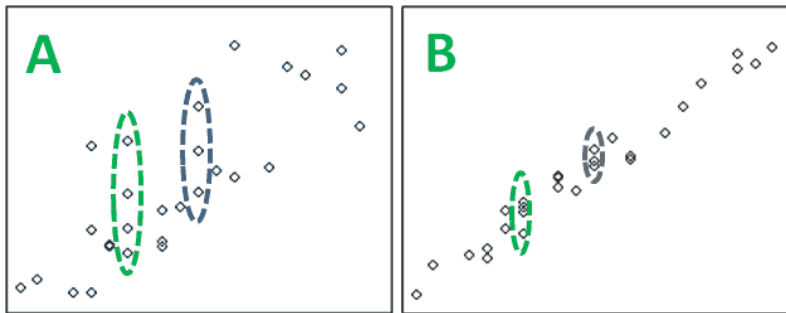


Interpreting scatterplots - Strength of the Relationship

The strength of the association between the factors describes how closely the points appear to follow a clear form or pattern

Scatterplots A and B show the age in months vs length in centimeters of **baby koala bears at two different zoos** (A and B, respectively).

The association between age in months and length in centimeters is more strongly associated when looking at the data from **Zoo B** than the data from **Zoo A**.



Scatterplots – Upsides and Downsides

Advantages:

Visual Relationship: Clearly shows the relationship (if there is any) between two variables.

Identifies Patterns: Helps in identifying trends, clusters, and outliers.

Simple and Intuitive: Easy to create and interpret, even for beginners.

Disadvantages:

Limited Variables: Only displays two variables at a time, making it difficult to analyze more complex relationships.

Overplotting: With large datasets, points can overlap, making it hard to see individual data points.

No Causation Indicated: While it shows correlation, it doesn't imply causation between variables.

Common Errors in Scatterplot and Solution

a. Not Labeling Axes Properly

Missing or unclear labels for the X and Y axes. It causes confusion about which variables are being compared.

Solution: Always include descriptive axis labels with units of measurement.

Example: Instead of "X" and "Y," use "Study Hours (per week)" and "Exam Scores."

b. Mis-assigning Variables

Placing the dependent variable on the X-axis and the independent variable on the Y-axis. It leads to misinterpretation of relationships.

Solution: Follow the convention: Explanatory (independent) variable on the X-axis and response (dependent) variable on the Y-axis.

Tip: Think about which variable is causing or influencing the other.

c. Overplotting with Large Datasets

Plotting too many data points, causing overlaps that obscure patterns. It masks trends, clusters, and outliers.

Solution: Use smaller point sizes or transparency (alpha blending) to make overlapping points visible.

Correlation

Correlation

Correlation (denoted as r) or the correlation coefficient is a measure of the strength and direction of a linear relationship between two quantitative variables in a sample.

The correlation (or the sample correlation coefficient) between two variables x and y can be computed using:

$$r = \frac{1}{n-1} \sum_{i=1}^n \left(\frac{x_i - \bar{x}}{s_x} \right) \left(\frac{y_i - \bar{y}}{s_y} \right)$$

- ▷ $(x_i - \bar{x})/s_x$ gives the number of standard deviations from the mean that the i th observation of x is.
- ▷ The sample correlation is an average of the product of the standardized x and y data points.

Properties of Correlation

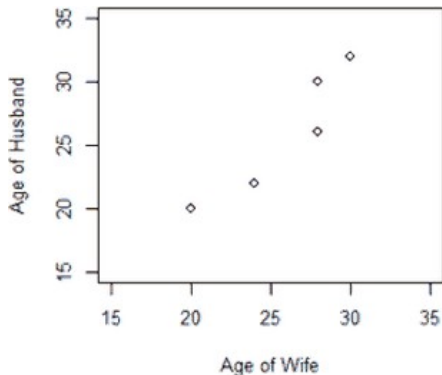
1. The **correlation** takes on **values between -1 and +1**.
2. The correlation **between variables x and y** is **the same as the correlation** between variables **y and x**.
3. Correlations can be computed between paired values of two **quantitative variables**.
4. The correlation coefficient **does not have units** and it is **independent of unit** of measure of variables x and y.
5. **Correlation measures the strength** of a linear relationship only.
Correlation should **not be used to describe a curved** relationship - even if the association is strong.
6. **Outliers affect correlation**. Correlation in the presence of outliers should be interpreted with caution.

An example - Correlation

We have data about age of 5 couples.

Calculate the sample correlation between the ages of husbands and wives.

Couple	Age of Wife	Age of Husband
1	20	20
2	30	32
3	24	22
4	28	26
5	28	30
Sample mean	26	26
Sample standard deviation	4.0	5.1

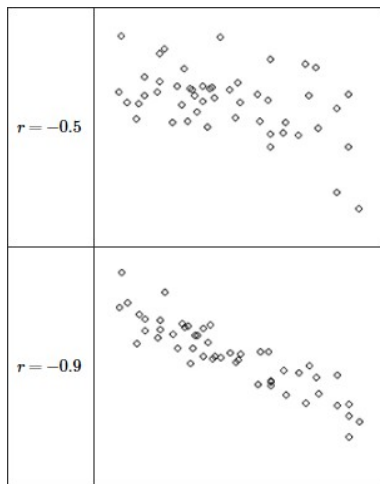
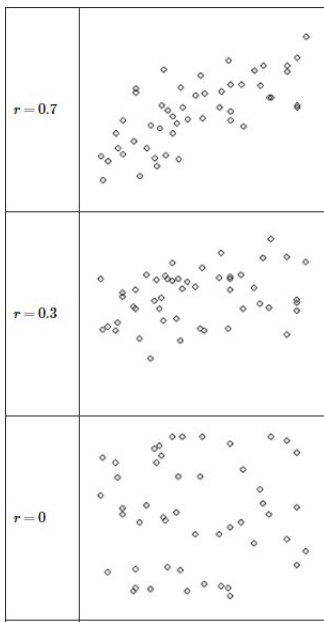


An Example - Correlation

Couple	Age of Wife	Age of Husband	Standardized Age of Wife $\left(\frac{x_i - \bar{x}}{s_x}\right)$	Standardized Age of Husband $\left(\frac{y_i - \bar{y}}{s_y}\right)$	$\left(\frac{x_i - \bar{x}}{s_x}\right) \left(\frac{y_i - \bar{y}}{s_y}\right)$
1	20	20	$\frac{x_i - \bar{x}}{s_x} = \frac{20 - 26}{4.0} = \frac{-6}{4}$	$\frac{y_i - \bar{y}}{s_y} = \frac{20 - 26}{5.1} = \frac{-6}{5.1}$	$\left(\frac{-6}{4}\right) \left(\frac{-6}{5.1}\right) = \frac{36}{20.4}$
2	30	32	$\frac{30 - 26}{4.0} = \frac{4}{4} = 1$	$\frac{32 - 26}{5.1} = \frac{6}{5.1}$	$\left(\frac{4}{4}\right) \left(\frac{6}{5.1}\right) = \frac{24}{20.4}$
3	24	22	$\frac{24 - 26}{4.0} = \frac{-2}{4}$	$\frac{22 - 26}{5.1} = \frac{-4}{5.1}$	$\left(\frac{-2}{4}\right) \left(\frac{-4}{5.1}\right) = \frac{8}{20.4}$
4	28	26	$\frac{28 - 26}{4.0} = \frac{2}{4}$	$\frac{26 - 26}{5.1} = \frac{0}{5.1}$	$\left(\frac{2}{4}\right) \left(\frac{0}{5.1}\right) = \frac{0}{20.4}$
5	28	30	$\frac{28 - 26}{4.0} = \frac{2}{4}$	$\frac{30 - 26}{5.1} = \frac{4}{5.1}$	$\left(\frac{2}{4}\right) \left(\frac{4}{5.1}\right) = \frac{8}{20.4}$

$$\begin{aligned} r &= \frac{1}{n-1} \sum_{i=1}^n \left(\frac{x_i - \bar{x}}{s_x} \right) \left(\frac{y_i - \bar{y}}{s_y} \right) \\ &= \frac{1}{5-1} \left(\frac{36}{20.4} + \frac{24}{20.4} + \frac{8}{20.4} + \frac{0}{20.4} + \frac{8}{20.4} \right) \\ &= \frac{1}{4} \left(\frac{76}{20.4} \right) \\ &\approx 0.93 \end{aligned}$$

Correlation coefficients

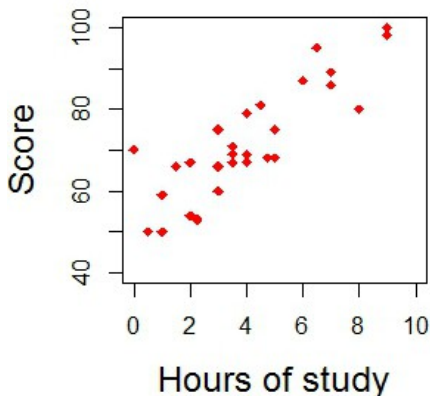


R function cor()

Use **cor()** to calculate sample correlation coefficient

```
> cor(data$explanatoryvariable, data$responsevariable)

# calculate sample correlation
> cor(study.hours, score)
> cor(score, study.hours)
0.8835101
```



Inference about population correlation coefficient

- ▷ Use sample data to make conclusions about **the population correlation coefficient**.
- ▷ **The Sample correlation, r , is a point estimate for the population correlation coefficient, ρ .**
- ▷ Formal tests of hypotheses concerning ρ seek to determine **whether there is a linear association between the variables in the population**.

$H_0 : \rho = 0$ (there is no linear association)

$H_1 : \rho \neq 0$ (there is a linear association)

Inference about population correlation coefficient

$H_0 : \rho = 0$ (there is no linear association)

$H_1 : \rho \neq 0$ (there is a linear association)

We use the test statistic:

$$t = r \sqrt{\frac{n-2}{1-r^2}}$$

which follows a **t-distribution with $n-2$ degrees of freedom under H_0 .**

- ▷ The decision rule for a two-sided level α test is:

Reject $H_0 : \rho = 0$ if $t \geq t_{n-2, \frac{\alpha}{2}}$ or if $t \leq -t_{n-2, \frac{\alpha}{2}}$

Otherwise do not reject $H_0 : \rho = 0$

- ▷ $t_{n-2, \frac{\alpha}{2}}$ is the value from the t-distribution table with $n-2$ degrees of freedom and associated with a right hand tail probability of $\alpha/2$.

An Example - Inference

Is there a linear relationship between hours of study and exam score?

Using the data we collected on the 31 students, we can test the hypothesis that $H_0 : \rho = 0$ (no linear association) versus $H_1 : \rho \neq 0$ (linear association).

1. Set up the hypotheses and select the alpha level

$H_0 : \rho = 0$ (there is no linear association)

$H_1 : \rho \neq 0$ (there is a linear association)

$\alpha = 0.05$

2. Select the appropriate test statistic

$$t = r \sqrt{\frac{n - 2}{1 - r^2}}$$

An Example - Inference

3. State the decision rule

Decision Rule: Reject H_0 if $p\text{-value} < \alpha$.

Otherwise do not reject H_0 .

OR

Determine the appropriate value from the t-distribution table with $n - 2 = 31 - 2 = \underline{29}$ degrees of freedom and associated with a right hand tail probability of $\frac{\alpha}{2} = 0.025$

Using the table, $t_{n-2, \frac{\alpha}{2}} = t_{29, 0.025} = 2.045$

Using R

```
> qt(0.975, df=29) = 2.04523
```

Decision Rule: Reject H_0 if $t \geq 2.045$ or if $t \leq -2.045$ ($|t| \geq 2.045$).

Otherwise, do not reject H_0

4. Compute the test statistic

$$t = r\sqrt{\frac{n-2}{1-r^2}} = 0.8835\sqrt{\frac{31-2}{1-0.8835^2}} \approx 10.16$$

5. Conclusion

- ▷ Reject H_0 since $10.16 \geq 2.045$.
- ▷ We have significant evidence at the $\alpha = 0.05$ level that $\rho \neq 0$.
- ▷ There is evidence of a significant linear association between study time and exam score.
- ▷ The sample correlation coefficient is 0.8835 indicating a strong positive association between study time and exam score. The positive correlation between these factors indicates that as study time increases, exam scores increase.

An Example - Inference

Advantages

- Quantitative Measure:** Provides a specific value indicating the strength and direction of the relationship.
- Easy Comparison:** Correlation coefficients allow for easy comparison between different pairs of variables.
- Foundation for Further Analysis:** Serves as a basis for more advanced statistical analyses.

Disadvantages

- Only Linear Relationships:** Correlation measures only linear relationships and may miss non-linear relationships.
- Sensitive to Outliers:** Outliers can significantly affect the correlation coefficient, potentially misleading the analysis.
- Does Not Imply Causation:** Similar to scatter plots, correlation does not indicate causation between variables.

Real-World Examples

- 1.**Temperature vs. Ice Cream Sales:** Higher temperatures might correlate with increased ice cream sales.
- 2.**Years of Experience vs. Salary:** Typically, more experienced employees may have higher salaries.
- 3.**Exercise Frequency vs. Health Indicators:** More frequent exercise might correlate with better health metrics like lower blood pressure.

R function cor.test()

Use cor.test() to perform testing

```
> cor.test(data$explanatoryvariable, data$responsevariable,
           alternative=[alternative], method=[method], conf.level=[confidence
           level])
```

[alternative] = **"two.sided"**, **"less"** (corresponds to negative association), or **"greater"** (corresponds to positive association)

[method] = **"pearson"**, **"kendall"**, or **"spearman"**

"pearson", the test statistic is based on Pearson's product moment correlation coefficient and follows a t distribution with $df = n-2$ (samples independent normal distributions).

If method is **"kendall"** or **"spearman"**, Kendall's tau or Spearman's rho statistic is used to estimate a rank-based measure of association. These might be used if the data is not normal.

Correlation: Upsides and Downsides

Simplicity and Ease of Use

Easy Calculation: Simple to compute with tools like Excel or calculators.

Clear Interpretation: Provides a single value indicating the relationship's strength and direction.

Quantifies Relationship

Strength: Measures how closely two variables move together.

Direction: Indicates whether the relationship is positive or negative.

Facilitates Comparison

Multiple Pairs: Allows comparison of relationships between different variable pairs.

Benchmarking: Helps identify which variables are more strongly related.

Foundation for Further Analysis

Regression Analysis: Serves as a preliminary step for more complex analyses.

Predictive Modeling: Assists in selecting potential predictor variables.

Correlation: Upsides and Downsides

Does Not Imply Causation

Misinterpretation Risk: High correlation doesn't mean one variable causes the other.

Third Variables: External factors may influence both variables, creating spurious relationships.

Sensitive to Outliers

Distortion: Extreme values can significantly affect the correlation coefficient.

Misleading Results: Outliers may create the illusion of a stronger or weaker relationship.

Only Measures Linear Relationships

Limited Scope: Fails to capture non-linear relationships between variables.

Overlooked Patterns: Important patterns may remain undetected if they aren't linear.

Affected by Data Range

Range Dependency: Correlation can vary based on the range of data analyzed.

Inconsistent Results: Different data ranges can lead to different correlation strengths.

Common Errors in Correlation and Solution

a. Confusing Correlation with Causation

Error: Believing that a correlation between two variables means one causes the other. It leads to incorrect conclusions and potentially poor decision-making.

Solution: Emphasize the principle: "Correlation does not imply causation."

b. Ignoring the Direction of Correlation

Error: Not noting whether the correlation is positive or negative. It misunderstands how variables relate to each other.

Solution: Always check the sign of the correlation coefficient (r). Visual Indicators: Use color-coding or arrows in examples to highlight the direction (positive vs. negative).

c. Not Recognizing Non-Linear Relationships

Error: Assuming that a correlation coefficient near zero means no relationship. It misses meaningful non-linear relationships.

Solution: Combine Visuals and Statistics: Always pair the correlation coefficient with a scatter plot to visually inspect for non-linear patterns. Teach Non-Linear Measures: Introduce alternative measures or transformations for non-linear data.

Simple Linear Regression

Simple Linear Regression (SLR)

If a linear relationship exists, **we can model** the nature of the relationship between the variables using **simple linear regression (SLR)**.

Using SLR, we assert a **straight line on the scatterplot** that represents the best fitting line to the data that captures the pattern of the relationship.

- ▷ An explanatory variable.
- ▷ A response variable

Simple Linear Regression (SLR)

We build a data model that allows us to:

- ▷ **quantify the relationship between the response variable and the explanatory variable**
- ▷ **predict the response of a new observation with a given value for x or what the average response is for observations with a specific value for the explanatory variable**

Simple Linear Regression (SLR)

The equation for the simple linear regression line is given by

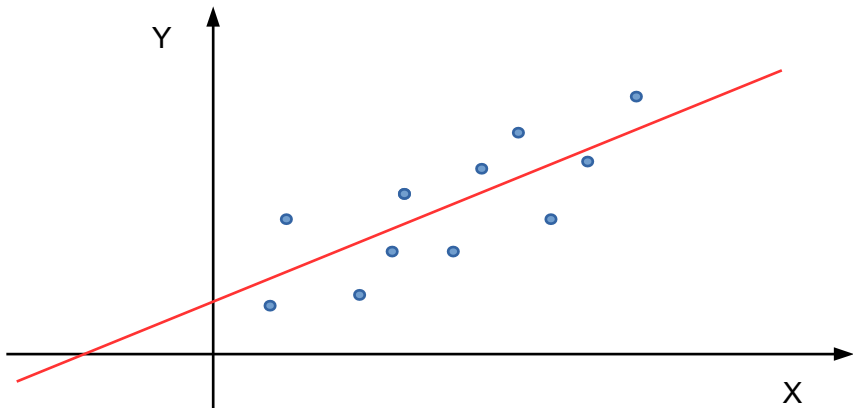
$$y = \beta_0 + \beta_1 x$$

- ▷ y is the response or dependent variable
- ▷ x is the explanatory or independent variable
- ▷ β_0 is the intercept (the value of y when $x = 0$)
- ▷ β_1 is the slope (the expected change in y for each one-unit change in x)

Simple Linear Regression (SLR)

The equation for the simple linear regression line is given by

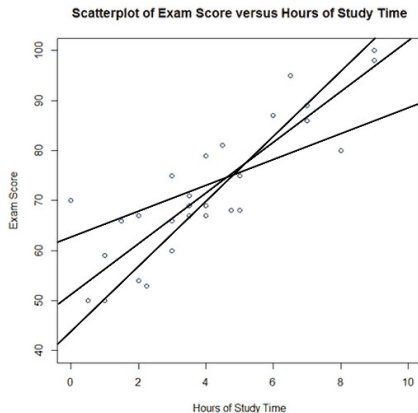
$$y = \beta_0 + \beta_1 x$$



How to find the regression line that best fits the data

There are several ways to find parameters of the line. The most common way is to **minimize the sum of the squares** of the distances between the points and the regression line.

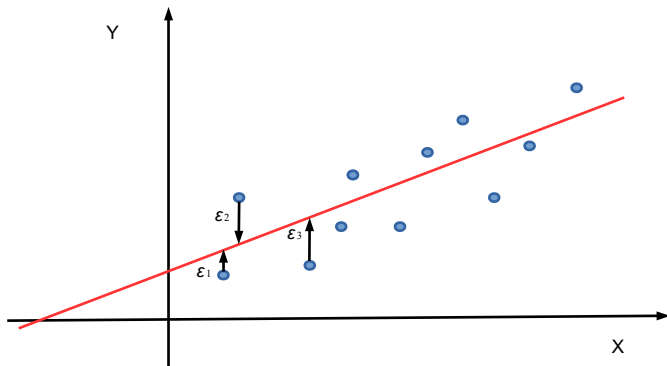
This approach is called the **least-squares method**. We want to minimize the vertical distance between each of the points and the regression line.



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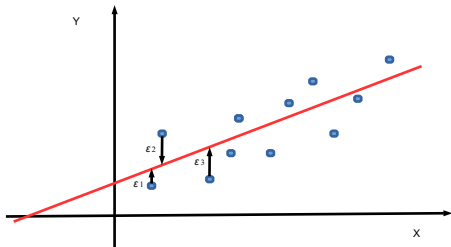
How to find the regression line that best fits the data

$$y_i = \beta_0 + \beta_1 x_i + \epsilon_i$$

$$\epsilon_i = y_i - (\beta_0 + \beta_1 x_i)$$

$$f(\beta_0, \beta_1) = \sum_{i=1}^n \left(y_i - (\beta_0 + \beta_1 x_i) \right)^2$$

Loss function, l2 norm



How to find the regression line that best fits the data

$$y_i = \beta_0 + \beta_1 x_i + \epsilon$$

$$\text{Best Fit Slope: } \hat{\beta}_1 = \frac{n \sum_{i=1}^n x_i y_i - \sum_{i=1}^n x_i \sum_{i=1}^n y_i}{n \sum_{i=1}^n x_i^2 - \left(\sum_{i=1}^n x_i \right)^2}$$

$$\text{Best Fit Y-Intercept: } \hat{\beta}_0 = \frac{\sum_{i=1}^n x_i^2 \sum_{i=1}^n y_i - \sum_{i=1}^n x_i \sum_{i=1}^n x_i y_i}{n \sum_{i=1}^n x_i^2 - \left(\sum_{i=1}^n x_i \right)^2}$$

Equation for the least- squares regression line

The equation for the simple linear regression line is given by

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x$$

In the least-squares regression, the estimates of β_0 and β_1 are:

$$\hat{\beta}_1 = r \frac{s_y}{s_x}$$

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$$

r is correlation coefficient

s_x the sample standard deviation of x , s_y is SD of y

$\bar{x}$ sample mean of x , and $\bar{y}$ sample mean of y

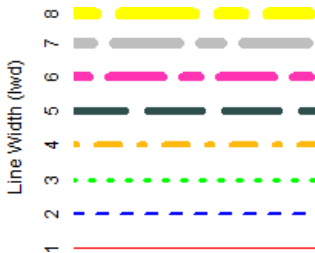
R Function - Simple linear regression

```
> lm(data$responsevariable ~ data$explanatory)
> abline(a=intercept, b=slope)

# create a model object
> m <- lm(score ~ study.hours)

# Add regression line to the scatterplot
> abline(m, lty=3, col="blue")
```

http://www.cookbook-r.com/Graphs/Shapes_and_line_types



An example - the least-squares regression line

National Unemployment Male vs. Female Reference: Statistical Abstract of the United States

```
> unemployment <- read.csv("national_unemployment_rate.csv")
> attach(unemployment)

> xbar <- mean(male.unemployment.rate)
> sx <- sd(male.unemployment.rate)
> ybar <- mean(female.unemployment.rate)
> sy <- sd(female.unemployment.rate)
> r <- cor(male.unemployment.rate, female.unemployment.rate)
> beta1 <- r*sy/sx
> beta0 <- ybar - beta1*xbar
```

$$\hat{\beta}_1 = 0.69$$

$$\hat{\beta}_0 = 1.43$$

$$\hat{y} = 1.43 + 0.69x$$

When $x = \bar{x} = 5.95$, $\hat{y} = \bar{y} = 5.57$